From 500 to a General Formula: Exploring a Floor Identity

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1 THE PROBLEM AND A CHANGE OF PERSPECTIVE

The original problem. Prove that there exists a real number w such that

$$\boxed{\sum_{k=0}^{500} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \lfloor wx \rfloor \qquad \text{for every } x \in \mathbb{R}.} \tag{1}$$

The number 500 makes the sum look formidable. But is there anything mathematically special about 500? The summands follow the same pattern, regardless of where we stop. This suggests that 500 may be an incidental feature of the question rather than an essential one.

Instead of trying to handle all 501 summands at once, let us replace 500 by a nonnegative integer m. The unknown number may depend on m, so we denote it by w_m. We are led to the more general problem: for each $m \in \mathbb{Z}_{\geqslant 0}$, find a real number w_m such that

$$\sum_{k=0}^{m} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \lfloor w_m x \rfloor \qquad \text{for every } x \in \mathbb{R}. \tag{2}$$

This change is useful precisely because it allows us to start with $m = 0$, then $m = 1$, and look for a pattern. The generalized problem may be easier to understand than the original numerical instance.

Since each identity must hold for *every* real x, we can choose test values of x that simplify the floor expressions. Such tests may suggest the value of w_m; we must then verify that it works for all real x.

2 EXPLORING THE GENERALIZED PROBLEM

2.1 The case $m = 0$

Equation (2) becomes

$$\left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor - \lfloor w_0 x \rfloor . \tag{3}$$

STEP 1: FIND A CANDIDATE. Take $x = 2n-1$ with $n \in \mathbb{Z}$. For the argument below, we choose $n \geqslant 1$, so that $2n-1 > 0$. Then

$$\left\lfloor \frac{(2n-1)+1}{2} \right\rfloor = n, \qquad \lfloor 2n-1 \rfloor = 2n-1.$$

If (3) holds for all x, it follows that

$$n = 2n-1-\lfloor w_0(2n-1) \rfloor, \qquad \text{so} \qquad \lfloor w_0(2n-1) \rfloor = n-1.$$

By the definition of the floor function,

$$n-1 \leqslant w_0(2n-1) < n.$$

By our choice $n \geqslant 1$, we have $2n-1 > 0$, and hence may divide the inequalities by $2n-1$ without reversing their directions:

$$\frac{n-1}{2n-1} \leqslant w_0 < \frac{n}{2n-1}.$$

The equality was assumed to hold for every real x, so these inequalities hold for every positive integer n. Both bounds tend to $\frac{1}{2}$ as $n \to \infty$. Therefore any w_0 satisfying the identity must equal

$$\boxed{w_0 = \frac{1}{2}.}$$

This is a *necessary condition*: so far we have not proved that it works for every real x.

STEP 2: VERIFY THE CANDIDATE. We recall *Hermite's floor identity*:

$$\sum_{r=0}^{N-1} \left\lfloor y + \frac{r}{N} \right\rfloor = \lfloor Ny \rfloor, \qquad y \in \mathbb{R}, \quad N \in \mathbb{Z}_{>0}. \tag{4}$$

For $N = 2$, this reads

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor. \tag{5}$$

Taking $y = x/2$ gives

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

Hence (3) holds for every real x with $w_0 = \frac{1}{2}$.

2.2 The case $m = 1$

Now the desired identity is

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \lfloor x \rfloor - \lfloor w_1 x \rfloor. \tag{6}$$

We follow the same two-step strategy.

STEP 1: FIND A CANDIDATE. As in the first case, take $x = 2n - 1$ with $n \in \mathbb{Z}$. To simplify the second floor term as well, choose $n = 2s$ with $s \in \mathbb{Z}_{>0}$. Thus our test values are

$$x = 4s - 1 > 0.$$

Then

$$\left\lfloor \frac{x+1}{2} \right\rfloor = 2s, \qquad \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor s + \frac{1}{4} \right\rfloor = s.$$

Thus (6) implies

$$3s = 4s - 1 - \lfloor (4s-1)w_1 \rfloor, \qquad \text{or} \qquad \lfloor (4s-1)w_1 \rfloor = s - 1.$$

By the definition of the floor function,

$$s - 1 \leqslant (4s-1)w_1 < s.$$

Our choice $s \geqslant 1$ ensures $4s - 1 > 0$. Dividing by this positive number gives

$$\frac{s-1}{4s-1} \leqslant w_1 < \frac{s}{4s-1}.$$

Because the assumed identity holds for all real x, these inequalities hold for every $s \in \mathbb{Z}_{>0}$. Letting $s \to \infty$, we obtain the necessary value

$$\boxed{w_1 = \frac{1}{4}.}$$

STEP 2: VERIFY THE CANDIDATE. Apply (5) twice. With $y = x/4$,

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor \frac{x}{2} \right\rfloor. \tag{7}$$

With $y = x/2$,

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor. \tag{8}$$

Adding these equalities and cancelling the common term $\lfloor x/2 \rfloor$ gives

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

This proves (6) for every $x \in \mathbb{R}$ with $w_1 = \frac{1}{4}$.

What have we discovered? The first two cases suggest the pattern

$$w_0 = \frac{1}{2}, \qquad w_1 = \frac{1}{4}, \qquad \text{and perhaps} \qquad w_m = \frac{1}{2^{m+1}}.$$

The cancellation in the second case is especially revealing: the identity at one scale links to the identity at the next scale. Can we repeat this cancellation for arbitrary m?

2.3 The general case

We now prove that the proposed formula works for every nonnegative integer m.

Theorem 2.1. *For every $m \in \mathbb{Z}_{\geqslant 0}$ and every $x \in \mathbb{R}$,*

$$\boxed{\sum_{k=0}^{m} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor + \left\lfloor \frac{x}{2^{m+1}} \right\rfloor = \lfloor x \rfloor .} \tag{9}$$

In particular, $w_m = 2^{-(m+1)}$ satisfies (2).

Proof. The key is again the $N = 2$ case of Hermite's identity,

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .$$

Use it successively with

$$y = \frac{x}{2^{m+1}},\ \frac{x}{2^m},\ \ldots,\ \frac{x}{4},\ \frac{x}{2}.$$

This gives

$$\begin{aligned}
\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor &= \left\lfloor \frac{x}{2^m} \right\rfloor , \\
\left\lfloor \frac{x}{2^m} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor &= \left\lfloor \frac{x}{2^{m-1}} \right\rfloor , \\
&\vdots \\
\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor &= \left\lfloor \frac{x}{2} \right\rfloor , \\
\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor &= \lfloor x \rfloor .
\end{aligned}$$

Add all these equations. Every intermediate term $\left\lfloor x/2^j \right\rfloor$, $1 \leqslant j \leqslant m$, appears once on each side and cancels. What remains is

$$\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor + \cdots + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor ,$$

which is exactly (9). □

Returning to the original problem, we take $m = 500$. Thus the required identity holds for every $x \in \mathbb{R}$ with

$$\boxed{w = \frac{1}{2^{501}}.}$$

3 REFLECTION: FROM A SPECIFIC PROBLEM TO A GENERAL PROOF

The original problem involved the number 500. But why 500? There seemed to be nothing special about this number. This led us to replace 500 by a general nonnegative integer m, and to let the unknown number depend on m, writing it as w_m.

This is an important step in mathematical thinking: *generalizing a problem can sometimes make it easier to solve.* Instead of working directly with a

sum containing 501 terms, we turned a single problem into a family of related problems.

At first sight, replacing 500 by an arbitrary m may seem to make the problem more difficult. However, it gives us something important: we can now investigate simpler cases.

We started with $m = 0$ and $m = 1$. In each case, we chose special values of x to find a necessary condition on w_m. This suggested the pattern

$$w_0 = \frac{1}{2}, \qquad w_1 = \frac{1}{4}, \qquad w_m = \frac{1}{2^{m+1}}.$$

But discovering a pattern is not the same as proving it. The special values of x helped us find possible values of w_m, but they did not establish the required equality for every real number x. For this, we used Hermite's floor identity.

Our reasoning therefore moved in two directions: we first generalized the original problem, and then studied its simplest cases to discover a possible general structure. Finally, we returned to the general problem and proved the conjectured formula.

This illustrates an important feature of mathematical problem-solving. We do not always have to solve a problem exactly as it is presented. We may first change the question, look for patterns in simpler cases, and then use rigorous reasoning to establish a general result.

A A PROOF OF HERMITE'S FLOOR IDENTITY

We prove that, for every $y \in \mathbb{R}$,

$$\boxed{\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .}$$

Proof. Let $n = \lfloor y \rfloor$. Then we can write

$$y = n + r, \qquad 0 \leqslant r < 1.$$

Hence

$$\lfloor y \rfloor = n$$

and

$$\left\lfloor y + \frac{1}{2} \right\rfloor = n + \left\lfloor r + \frac{1}{2} \right\rfloor .$$

On the other hand,

$$\lfloor 2y \rfloor = 2n + \lfloor 2r \rfloor .$$

Since $0 \leqslant r < 1$, we distinguish two cases:

	$\left\lfloor r + \frac{1}{2} \right\rfloor$	$\lfloor 2r \rfloor$
$0 \leqslant r < \frac{1}{2}$	0	0
$\frac{1}{2} \leqslant r < 1$	1	1

Thus,

$$\left\lfloor r + \frac{1}{2} \right\rfloor = \lfloor 2r \rfloor ,$$

which gives

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .$$

□